

ADR-002 — Showdown is the authority, MEDICHAM is the runtime

Version 3.50.0 · 2026-08-06 · Status: ACCEPTED · Supersedes the migration half of [ADR-001](#), which stands as a record and is not deleted.

Context

ADR-001 decided that `data/mods/champions/` becomes the authoritative rules engine, driven through the Showdown simulator API, and that `engine/medicham2-browser.js` becomes a lookup over precomputed matchup tables with the hand-written engine as fallback only.

The forcing evidence was severe and correctly read: the official simulator and the hand-written engine disagreed by **31.1 percentage points** of win probability on identical teams under identical policy, flipping the favourite in **3 of 8** matchups. Every hand-fix made that day moved the engine **4.35 points**. ADR-001's conclusion follows from those numbers:

"The remaining disagreement is seven times larger than everything we fixed. Fixing this engine by hand was never going to converge."

Migration steps 5–8 were never executed. The opposite happened. `medicham2-browser.js` is now the property of a standing division whose one number — mechanics live — may never fall, and it is the leaf of the live search player. That is ADR-001's explicitly rejected alternative, adopted in practice, with nothing recording the change. The repository's stated authority and its actual authority have been different files for over a week.

This ADR does not discover that drift. It ends it, by deciding which of the two is right.

Decision

Showdown is the AUTHORITY. MEDICHAM is the RUNTIME. They are different jobs and both are kept.

- **Showdown decides what is correct.** This is not a preference, it is wired into the tests: `tests/test-engine-diff.js` and `tests/test-interaction-matrix.js` both treat the official engine's answer as truth and score any disagreement as a MEDICHAM bug, never the reverse. The matrix authors no expected outcome at all — it asks the reference engine. It extends past mechanics to the format itself: what is banned and what is legal is read out of the mod (`isNonstandard: 'Past'`), never remembered.
- **MEDICHAM executes.** It is what actually runs in rollouts, self-play and the live bot.
- **The gap between them is measured, published, and ratcheted.** It is not assumed to be zero.

Why the decision reverses, stated as evidence rather than preference

1. The premise was falsified by events. ADR-001 held that hand-fixing would never converge. It converged. Read at the moment of the decision, from artifacts, not prose:

instrument	agreement with the official engine
data/engine-diff.json (damage, seed 20260804)	20,000 / 20,000 , clean. (This row read 150 / 150 while the artifact happened to hold a short run; the sample size is an argument to the harness, not a property of the engine, so it is re-read from the artifact rather than quoted from a previous session — ROADMAP #161, 2026-08-11.) (This read 149 / 150 and called the residual "a documented harness-layer boundary, both engines right". That was wrong and is retracted, 2026-08-10. The residual was chesnaught woodhammer -> mimikyu , and it was a real engine defect: dmgRange ignored an intact Disguise and priced the click at 120-130 where the authority says 0. Both engines were NOT right. The claim survived because the battle LOOP handles Disguise correctly, so the mechanic looked implemented — see docs/ENGINE.md .)
data/interaction-matrix.json (carrier × reactor)	1,640 / 1,640 — 100.0% of 2,253 staged, PLUS the artifact's own off_gate count of 30 disagreements in buckets the gate discards. (Two rows of this were still parting mid-queue; ROADMAP #161 closed them on 2026-08-11 and the row is re-read from the artifact rather than carried forward — the superseded pair of figures is not restated here, because a stale number printed beside a live one is what this whole check exists to catch.) The fixture regression this row carried on 2026-08-11 is diagnosed and fixed — see the note under this table.
data/mechanics-census.json	the live/probed count, with every miss carrying a written reason. Read it from the artifact — this figure moved four times in one night as the WIRE queue landed, and was hand-corrected three of them.

THE MATRIX ROW WAS A FIXTURE REGRESSION AND IT IS FIXED, 2026-08-11. This row briefly read a *lower* live count with a *higher* agreement rate, and the suspicion recorded here at the time — that the harness had stopped exercising the protect family rather than reclassifying it — was correct. The control carrier for a reactor-MOVE case asked a **usage-ranked** index whether a candidate carried the flag under test, so Flail (contact, and unplayed, therefore absent from that index) was chosen as the "flagless" control; both arms made contact, Spiky Shield chipped both, and the pairs reported INERT — unscored, and invisible in the rate. The membership question now has one implementation, engine/linkage_carrier.js , shared with the index builder. Full account and counts in docs/ENGINE.md .

(Those are a snapshot, and the census in particular moves — it read 202/211 four hours before this was written, 211/214 when it was checked, 216/219 after WIRES 119–122, 217/220 after WIRE 123, 218/221 after WIRE 128, 231/232 after WIRES 129–130 and 234/235 after WIRES 131–132. That is the point rather than a caveat: the number climbs because the instruments exist. Read it from the artifact, never from this table.)

What made the difference is not stubbornness. **The differential harness and the interaction matrix did not exist when ADR-001 was written.** Hand-fixing did not converge; instrumented fixing did. ADR-001 was right about the engine it could see and wrong about the engine that could be built once something was measuring it.

2. The migration is technically impossible for the thing we now build. ADR-001 accepted, openly, that the official simulator is **117× slower** (29 vs 3,401 battles/sec/core) and that live browser simulation would be lost. That cost was acceptable for a precomputed matchup table. It is not survivable under MILTANK, which decides by playing positions out thousands of times per turn. You cannot put a 117× slowdown beneath a rollout search. The project acquired a search player after ADR-001 was written, and that changes what the runtime has to be.

CORRECTED 2026-08-06 (3.59.0). *The ratio quoted in the paragraph above is not true, and this ADR's conclusion does not depend on it.* Re-measured on the same machine, the same four teams (derived from the store), 8-second runs at a 60-turn cap: MEDICHAM **13,041** turns/sec against champions_sim 's **523** — a ratio of **24.9x**, not 117x. On battles/sec the figures are 217 against 28, but the two engines were driven differently (MEDICHAM to its 60-turn cap, Showdown with choose('default') to a natural end), so **turns/sec is the comparable unit and battles/sec is not.** The prior figures are kept above rather than rewritten. The argument here needs only that the gap is

large enough to sit under a per-turn rollout search, and 24.9x is: MILTANK already needs **26 s against a 20 s budget** on one core, and 24.9x that is not a search, it is a forfeit. **What this correction does change is why ADR-001's decision was right.** It was justified with a number that is not true, and the honest statement is now the one ADR-003 makes: MEDICHAM exists so that per-turn re-solving is affordable, and **the engine work is justified if and only if search pays.** That is a falsifiable claim with a gate attached (ROADMAP #62), which the 117x never was.

3. Nothing about this weakens the oracle. Keeping MEDICHAM as the runtime does not promote it to the authority. Every disagreement remains MEDICHAM's fault by construction.

What we accept, and it is not small

Conformance is proven only where we have looked. The matrix stages **2,300 of a theoretical 8,795** carrier × reactor pairs — **26.2%**. The honest claim is *"the engine agrees with the official engine on the 2,300 pairs that ran"*, never *"the engine is correct"*. That coverage fraction **is** the answer to how much of the law we actually follow, and raising it is the entire point of the coverage programme (mutation → unified generator → registry → differential).

And a coverage instrument can inflate its own findings. The mutation tier reported 97 defect candidates in 3.49.0; the two largest were a deliberate doubles override (Light Screen) and a name-branched recoil (Life Orb), and neither is a defect. A mutation verdict says what MOVED and cannot see intent, so 3.49.1 grades every open operator A/B/C/D from a **parse of the frozen engine source** — nought of the 97 is class A — and ratchets class A alone. Same guard as above, pointed at ourselves: the classifier is calibrated on three cases decided by hand before it existed and refuses to publish if it cannot reproduce them.

The failure mode to watch is the quiet one. When a differential row goes red there are two ways to make it green: fix the engine, or fix the test. Both are sometimes correct — when the engine learned crits the residual went 1/400 → 5/400 and the *harness* was wrong. Do that carelessly a few times and the hand-written engine becomes the authority without anyone deciding. The standing guard: **a flagged row is still counted in the residual.** Flagging never moves the number.

The store is not public and the pin is a commit. Both carried over from ADR-001 and both stand.

Consequences

- `engine/medicham2-browser.js` stays a first-class, division-owned component. Its census may never fall.
- The precomputed matchup table is **not** built. ADR-001's steps 5–8 are formally withdrawn.
- `engine/champions_sim.js` stays the oracle, pinned by commit, resolved from a sibling checkout.
- Any claim about mechanics cites an instrument and its coverage, not the engine's self-report.
- **If the coverage programme stalls, this decision weakens.** The justification is the measured agreement above; it is not a licence, and it expires if the instruments stop being run.

Sources

Measured, from artifacts on disk 2026-08-05. `data/engine-diff.json` , `data/interaction-matrix.json` , `data/mechanics-census.json` . These are the figures this decision rests on and each traces to a file.

Quoted from ADR-001, not re-measured. The 31.1-point win-probability disagreement, the 4.35-point hand-fix delta and the 117× simulator timing appear nowhere in the artifacts above and must not be attributed to them. They are ADR-001's record of what was known in July, kept because a decision reversal is only legible beside the evidence it reverses. If any of them matters again it gets re-measured before it is quoted.

And one of the three has now been re-measured, which is why that sentence was written. The timing was re-run on 2026-08-06 and came back at **24.9x** on turns/sec, not 117x — see the correction note in point 2 above. It has no artifact under `data/` either; it is a benchmark of two engines on one machine at one moment, and this ADR states it as such rather than citing a file that does not exist. **Nothing about the 31.1-point or 4.35-point figures has changed**, and neither should be quoted until it has been through the same treatment.

(The doc-currency check flagged those three as figures a cited artifact does not contain. It was right to. They are historical quotes sitting in a document that also cites live artifacts, and the check cannot tell those apart from a paragraph — so the document says which is which instead.)

Revision, 3.43.0 — the matrix figures in this ADR moved, and the direction is uncomfortable

This ADR was written citing **1,514 of 8,506 pairs staged** and **899 / 899 — 100.0% agreement**. Both are now superseded: **1,675 of 8,676 staged, 1,027 / 1,031 — 99.6%**. Nothing about the decision changes, but the record has to say which way the number went and why.

The matrix never checked its own arithmetic. Once `theoretical = staged + dropped` was asserted per axis it failed immediately, on three separate faults (an understated denominator, a depth-cap off-by-one, and outcome buckets that were not a partition — see `CHANGELOG.md` 3.43.0). Closing it recovered 161 pairs the generator had been discarding without naming them, and among those pairs sit four the engine gets wrong.

So the 100.0% in the table above was an artefact of a denominator that dropped 5,090 pairs in silence. It is exactly the failure this ADR argues *against* — a conformance claim that reads as proof and is really a statement about where nobody looked. A 99.6% that knows its own denominator is the stronger evidence for keeping MEDICHAM, not weaker.